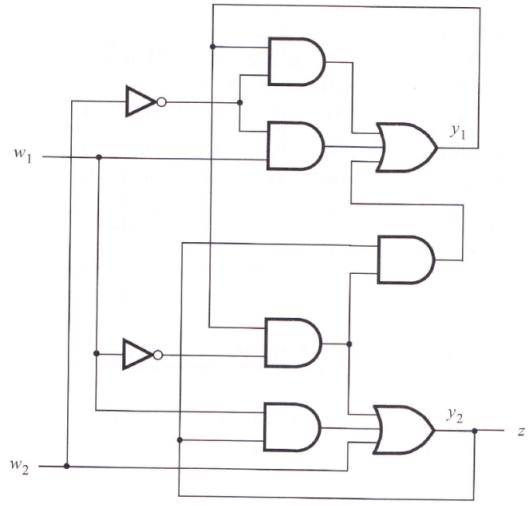
# CE2003: Tutorial 9

Q1. Determine the reduced state flow diagram from the flow diagram below.

| Present State | $w_2 w_1$ 00 | 01 | 11 | 10 | Output z |
|---|---|---|---|---|---|
| A | **A** | B | -- | C | 0 |
| B | A | **B** | F | -- | 0 |
| C | A | -- | F | **C** | 0 |
| D | **D** | E | -- | C | 1 |
| E | D | **E** | F | -- | 0 |
| F | -- | E | **F** | C | 1 |

Q2 Determine the flow table that describes the behaviour of the circuit below.




Q3 Reduce the flow table to one with just 4 states.

| Present State | $w_2 w_1$ 00 | 01 | 10 | 11 | Output z |
|---|---|---|---|---|---|
| A | **A** | F | C | -- | 0 |
| B | A | **B** | -- | H | 1 |
| C | G | -- | **C** | D | 0 |
| D | -- | F | -- | **D** | 1 |
| E | G | -- | **E** | D | 1 |
| F | -- | **F** | -- | K | 0 |
| G | **G** | B | J | -- | 0 |
| H | -- | L | E | **H** | 1 |
| J | G | -- | **J** | -- | 0 |
| K | -- | B | E | **K** | 1 |
| L | A | **L** | -- | K | 1 |

## Answers:

Remember

1. Both $S_i$ and $S_j$ have the same output condition
2. Both $S_i$ and $S_j$ have the same successor (next state for an input condition)
3. Both $S_i$ and $S_j$ are stable
4. The successor of $S_i$ and $S_j$ (or both) is unspecified

## Q1: This problem is very trivial

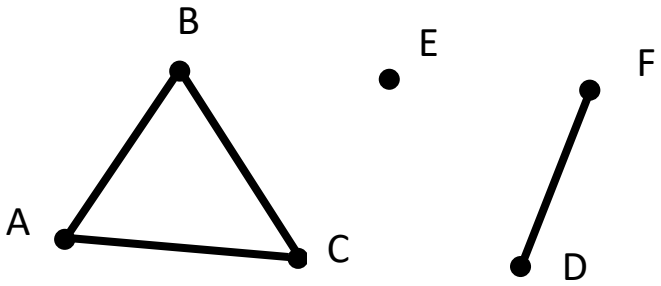
| Present State | $w_2 w_1$ 00 | 01 | 11 | 10 | Output z |
|---|---|---|---|---|---|
| A | **A** | B | -- | C | 0 |
| B | A | **B** | F | -- | 0 |
| C | A | -- | F | **C** | 0 |
| D | **D** | E | -- | C | 1 |
| E | D | **E** | F | -- | 0 |
| F | -- | E | **F** | C | 1 |

First partition based on output (item #1)

P1 = (ABCE) (DF)

P2 = ((AB)(AC)(BC)E) ((DF))

Then the merger diagram is:




Then an examination of the states shows that we can combine A, B and C into a single state (items #2 and #4). Don't make the mistake of trying to combine D, E and F into a single state. Note that E is in a different partition than D and F (due to a different output (z)). Can only combine D and F. The reduced flow diagram is:

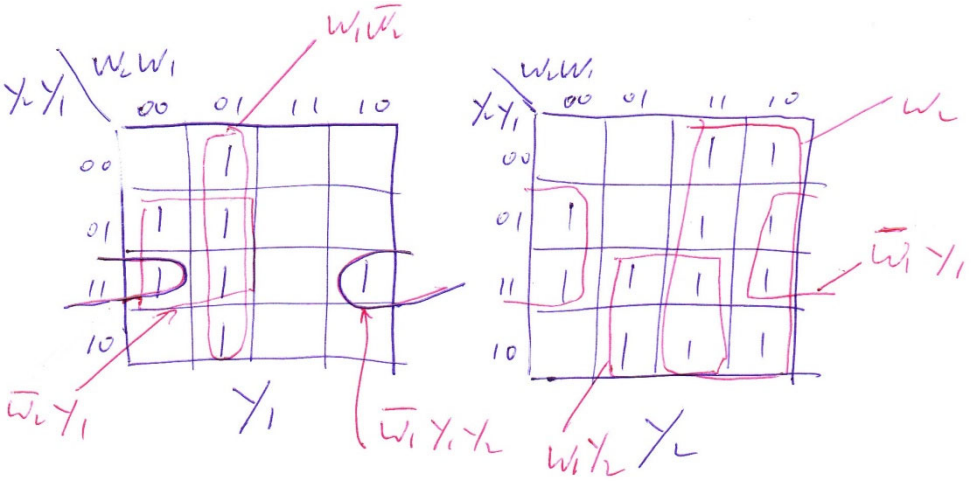
| Present State | $w_2 w_1$ 00 | 01 | 11 | 10 | Output z |
|---|---|---|---|---|---|
| A | **A** | **A** | D | **A** | 0 |
| D | **D** | E | **D** | A | 1 |
| E | D | **E** | D | -- | 0 |

**Q2:** We need to perform a reverse procedure to what we normally do. (This is easier as we do not need to worry about state reduction or state assignment)

First determine the next state and output expressions as:

$$Y_1 = w_1 w_2' + w_2' y_1 + w_1' y_1 y_2$$
$$Y_2 = w_2 + w_1' y_1 + w_1 y_2$$
$$z = y_2$$

Then we can start to determine the Excitation table, by firstly getting the K-maps




From the K-maps we can get the excitation table.

Note: make it easy by arranging the inputs and states in grey code format, that is directly as per the K-Map. Less likely to make mistakes, and it actually makes no difference. Also note the stable states.

| Present State $y_2y_1$ | $w_2$ $w_1$ 00 | 01 | 11 | 10 | Output z |
|---|---|---|---|---|---|
| | $Y_2Y_1$ | $Y_2Y_1$ | $Y_2Y_1$ | $Y_2Y_1$ | |
| 00 | **00** | 01 | 10 | 10 | 0 |
| 01 | 11 | **01** | 10 | 10 | 0 |
| 11 | **11** | **11** | 10 | **11** | 1 |
| 10 | 00 | 11 | **10** | **10** | 1 |

Then we can simply label states as: A = 00; B = 01; C = 11; and D = 10; to give the final flow table.

| Present State $y_2y_1$ | $w_2$ $w_1$ 00 | 01 | 11 | 10 | Output z |
|---|---|---|---|---|---|
| A | **A** | B | ~~D~~ | D | 0 |
| B | C | **B** | D | ~~D~~ | 0 |
| C | **C** | **C** | D | **C** | 1 |
| D | A | C | **D** | **D** | 1 |

Note the don't cares for states A & B, as 2 bits of the input are not allowed to change at the same time. You cannot transition from stable A to D with an input change 00 -> 11.

Q3: This is a bit more difficult than Q1 (and is probably too difficult for an examination question)

Firstly partition based on output condition (item #1)

P1 = (ACFGJ) (BDEHKL)

Secondly examine states for obvious equivalences

(A F) and (G J) are obvious combinations in group 1 partition 1 (items #2 and #4)

(D E) are similar in partition 1 (group 2) (by items #2 and #4)

P2 = ((AF) (CJ) (FJ) (GJ)) (B (DE) H K L)

So combine (AF), (GJ) and (DE) states and determine a new state flow diagram

| Present State | $w_2$ $w_1$ 00 | 01 | 10 | 11 | Output z |
|---|---|---|---|---|---|
| A | **A** | ~~F~~ **A** | C | -- K | 0 |
| B | A | **B** | -- | H | 1 |
| C | G | -- | **C** | D | 0 |
| D | -- G | ~~F~~ A | -- **D** | **D** | 1 |
| ~~E~~ | ~~G~~ | ~~--~~ | ~~**E**~~ | ~~D~~ | ~~1~~ |
| ~~F~~ | ~~--~~ | ~~**F**~~ | ~~--~~ | ~~K~~ | ~~0~~ |
| G | **G** | B | ~~J~~ **G** | -- | 0 |
| H | -- | L | E D | **H** | 1 |
| ~~J~~ | ~~G~~ | ~~--~~ | ~~**J**~~ | ~~--~~ | ~~0~~ |
| K | -- | B | ~~E~~ D | **K** | 1 |
| L | A | **L** | -- | K | 1 |

So the new state flow diagram is:

| Present State | $w_2$ $w_1$ 00 | 01 | 10 | 11 | Output z |
|---|---|---|---|---|---|
| A | **A** | **A** | C | K | 0 |
| B | A | **B** | -- | H | 1 |
| C | G | -- | **C** | D | 0 |
| D | G | A | **D** | **D** | 1 |
| G | **G** | B | **G** | -- | 0 |
| H | -- | L | D | **H** | 1 |
| K | -- | B | D | **K** | 1 |
| L | A | **L** | -- | K | 1 |

A quick look at the resulting table shows that (C G) can be combined. $W_2W_1$ = 10 has them both as there respective stable states (item #3). The other inputs are combined by (items #2 and #4).

| Present State | $w_2$ $w_1$ 00 | 01 | 10 | 11 | Output z |
|---|---|---|---|---|---|
| A | **A** | **A** | C | K | 0 |
| B | A | **B** | -- | H | 1 |
| C | **C** | B | **C** | D | 0 |
| D | C | A | **D** | **D** | 1 |
| H | -- | L | D | **H** | 1 |
| K | -- | B | D | **K** | 1 |
| L | A | **L** | -- | K | 1 |

Now it gets a bit harder as the obvious combinations are all done and now we must look at individual successors (next state for an input condition). But first let's get a partition based on output condition.

P1 = (A C) (B D H K L)

First try to combine (AC). To do this we should look at the successor for each input

The successors for (AC) are (AC); (AB); (CC); (KD) for $W_2W_1$ = 00; 01; 10; 11 respectively. If we look at the 2nd successor (AB), we see that they are in different output groups (ie A has output '0" and B has output '1'). So they cannot be identical and cannot be merged into a single state.

Next let's look at combinations in the second group in partition 1

Examining state D (for input condition $W_2W_1$ = 01) has the successor of D being A (in the z=0 group) while the other states (B H K L) have successors in the z=1 group. So D cannot be combined with anything (So states A C and D will be in the final minimised state flow diagram).

Now, lets look at the remaining 4 states {B, L, H and K}.

The successors for (BLHK) are (AA--); (**BL**LB); (--DD); (HK**HK**).
So B & L (both stable states) could be combined if H & K are compatible and vice versa.
That is, B-L is compatible if H-K is compatible, and H-K is compatible if B-L is compatible. Hence we can assume they are compatible and then we get a new partition

P2 = ((A) (C)) ((B L) (D) (H K)),

Thus we can group BL and HK into 2 states B and H. So we get a new state flow diagram as:

| Present State | $w_2 w_1$ 00 | 01 | 10 | 11 | Output z |
|---|---|---|---|---|---|
| A | **A** | **A** | C | H | 0 |
| B | A | **B** | -- | H | 1 |
| C | **C** | B | **C** | D | 0 |
| D | G | A | **D** | **D** | 1 |
| H | -- | B | D | **H** | 1 |

Lastly, we can see by inspection that B and H are identical states, which we would expect. So the final state flow diagram is:

| Present State | $w_2 w_1$ 00 | 01 | 10 | 11 | Output z |
|---|---|---|---|---|---|
| A | **A** | **A** | C | B | 0 |
| B | A | **B** | D | **B** | 1 |
| C | **C** | B | **C** | D | 0 |
| D | C | A | **D** | **D** | 1 |

Which we put into the k-map compatible form by swapping columns 3 and 4.

| Present State | $w_2 w_1$ 00 | 01 | 11 | 10 | Output z |
|---|---|---|---|---|---|
| A | **A** | **A** | B | C | 0 |
| B | A | **B** | **B** | D | 1 |
| C | **C** | B | D | **C** | 0 |
| D | C | A | **D** | **D** | 1 |

Note we do not swap rows C and D. there is no need at this point. We would only (possibly) do this when we look at state assignment.